



Course Prepared by
IITian & Data Analytics
Expert

DATA ANALYTICS CERTIFICATION PROGRAM

Explore Data. Visualize Trends. Drive Decisions.

SKILL IT Education presents a structured, practical career pathway into Data Analytics — taking learners step-by-step from Excel and SQL to exploratory analysis, visualization and business intelligence. Every concept is reinforced through hands-on labs, real-world datasets and industry-aligned analytics practices.

PROGRAM HIGHLIGHTS

- ✓ 9 Core Data Analytics Modules
- ✓ 5+ Hands-On Labs & Capstone Project
- ✓ Business Intelligence, Reporting & KPI Tracking
- ✓ 2-Month Industry Internship

PROGRAM DURATION

Total Duration	5 Months
Structured Learning	3 Months
Industry Internship	2 Months

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YOUR DATA ANALYTICS JOURNEY

Fundamentals

Excel

SQL

EDA

Visualization

BI Tools

Reporting

KPI Tracking

Capstone

WHY THIS MATTERS

Every business now runs on data, and organizations urgently need people who can turn numbers into decisions — not just build spreadsheets. Data analytics skills are among the fastest-growing and most in-demand across every industry today.



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Program Overview

Build Your Data Analytics Career

What This Program Covers

This program takes you step-by-step from practical Excel and SQL foundations to real analytics work — cleaning and exploring data, visualizing insights, building BI dashboards and reporting KPIs that drive decisions.

Why This Matters

Every business now runs on data, and organizations urgently need people who can turn numbers into decisions — not just build spreadsheets. Data analytics skills are among the fastest-growing roles in the job market today.

COURSE DURATION

3 Months

PROJECT LEARNING

Integrated

INTERNSHIP

2 Months

PROGRAM ROADMAP

- 1 Fundamentals of Data Analytics
- 2 Excel for Data Analysis
- 3 SQL for Data Analysis
- 4 Exploratory Data Analysis (EDA)
- 5 Data Visualization (Matplotlib / Seaborn)
- 6 Business Intelligence Tools (Power BI / Tableau)
- 7 Business Reporting & Dashboards
- 8 KPI Tracking & Business Metrics
- 9 Domain Knowledge + Capstone Projects

*From Numbers to Insight. From Reports to Decisions.
From Skills to Career.*

Module 1 — Fundamentals of Data Analytics

WHAT YOU WILL LEARN

- Introduction to data analytics & its business role
- Types of analytics: descriptive, diagnostic, predictive, prescriptive
- The data analytics lifecycle
- Data-driven decision making
- Data types, sources & collection methods
- Analytics tools landscape overview
- Setting up your analytics environment
- Structuring an analytics problem statement

TOOLS & TECHNOLOGIES

Excel

Google Sheets

Power BI

Tableau

CORE CONCEPTS

Descriptive

Diagnostic

Predictive

Prescriptive

Data Lifecycle

Decision Making

✓ LEARNING OUTCOME

Students build a clear foundation in what data analytics is, why it matters to business, and how the analytics lifecycle connects data to decisions.

Module 2 — Excel for Data Analysis

WHAT YOU WILL LEARN

- Excel formulas & functions for analysis
- Pivot tables & pivot charts
- Data cleaning with Excel
- VLOOKUP, XLOOKUP & INDEX-MATCH
- Conditional formatting & data validation
- What-if analysis & scenario tools
- Building dashboards in Excel
- Automating tasks with basic macros

TOOLS & TECHNOLOGIES

Excel

Google Sheets

Power Query

CORE CONCEPTS

Pivot Tables

VLOOKUP

XLOOKUP

Data Validation

Macros

Dashboards

✓ LEARNING OUTCOME

Students gain hands-on mastery of Excel as an analytics tool, from cleaning raw data to building interactive dashboards.

Module 3 — SQL for Data Analysis

WHAT YOU WILL LEARN

- Relational databases & SQL fundamentals
- SELECT, WHERE, ORDER BY & filtering data
- JOIN, GROUP BY & aggregate functions
- Subqueries & CTEs
- Window functions for analysis
- Writing queries for business reporting
- Database design basics
- Query optimization fundamentals

TOOLS & TECHNOLOGIES

SQL

MySQL / PostgreSQL

pgAdmin

SQL TOOLKIT

SELECT

JOIN

GROUP BY

Subqueries

Window Fn.

CTEs

✓ LEARNING OUTCOME

Students learn to write efficient SQL queries to extract and aggregate business data directly from relational databases.

Module 4 — Exploratory Data Analysis (EDA)

WHAT YOU WILL LEARN

- Univariate, bivariate & multivariate analysis
- Summary statistics & data profiling
- Identifying patterns, trends & anomalies
- Correlation analysis between variables
- Outlier detection techniques
- Handling missing & skewed data
- Hypothesis-driven data exploration
- Communicating early insights clearly

TOOLS & TECHNOLOGIES

Pandas

NumPy

Jupyter

Pandas Profiling

EDA WORKFLOW

Collect

Profile

Explore

Detect

Summarize

✓ LEARNING OUTCOME

Students learn to systematically explore business datasets to uncover patterns and generate the insights that guide every reporting decision that follows.

Module 5 — Data Visualization (Matplotlib / Seaborn)

WHAT YOU WILL LEARN

- Principles of effective data visualization
- Line, bar, scatter & histogram plots
- Matplotlib figure & axes customization
- Seaborn statistical plots & heatmaps
- Multi-panel & faceted visualizations
- Storytelling with data
- Choosing the right chart for the data
- Building simple visualization dashboards

TOOLS & TECHNOLOGIES

Matplotlib

Seaborn

Plotly

CHART TYPES

Line

Bar

Scatter

Histogram

Heatmap

Box Plot

✓ LEARNING OUTCOME

Students learn to turn raw analysis into clear, compelling visuals that communicate findings to business stakeholders.

Module 6 — Business Intelligence Tools (Power BI / Tableau)

WHAT YOU WILL LEARN

- BI concepts & dashboard design principles
- Connecting & importing data sources
- Building interactive dashboards & reports
- DAX basics in Power BI
- Calculated fields & filters in Tableau
- KPI tracking & drill-down analysis
- Publishing & sharing dashboards
- Presenting insights to stakeholders

TOOLS & TECHNOLOGIES

Power BI

Tableau

Excel

BI WORKFLOW

Connect

Model

Visualize

Publish

Present

✓ LEARNING OUTCOME

Students learn to build and present interactive business dashboards that turn data into decisions for real organizations.

Module 7 — Business Reporting & Dashboards

WHAT YOU WILL LEARN

- Structuring effective business reports
- Report automation & scheduling
- Designing dashboards for different audiences
- Executive summary writing
- Combining multiple data sources into one report
- Version control for recurring reports
- Report QA & accuracy checks
- Presenting reports to leadership

TOOLS & TECHNOLOGIES

Power BI

Excel

Google Data Studio

CORE CONCEPTS

Report Design

Automation

Exec. Summary

QA Checks

Presentation

✓ LEARNING OUTCOME

Students learn to design, automate and present recurring business reports that leadership can act on with confidence.

Module 8 — KPI Tracking & Business Metrics

WHAT YOU WILL LEARN

- Defining KPIs & business metrics
- Aligning metrics to business goals
- Building KPI dashboards & scorecards
- Tracking trends over time
- Setting benchmarks & targets
- Metric ownership & governance
- Common pitfalls in KPI design
- Communicating metric performance

TOOLS & TECHNOLOGIES

Power BI

Tableau

Excel

CORE CONCEPTS

KPI Design

Scorecards

Benchmarks

Trend Tracking

Governance

✓ LEARNING OUTCOME

Students learn to define, track and communicate the metrics that matter most to a business's performance.

Module 9 — Domain Knowledge + Capstone Projects

WHAT YOU WILL LEARN

- Applying analytics to real business domains
- Framing a business problem as an analytics project
- End-to-end capstone project execution
- Data collection through final presentation
- Working with stakeholders & requirements
- Presenting findings to a non-technical audience
- Building a portfolio-ready case study
- Preparing for analytics interviews

TOOLS & TECHNOLOGIES

Excel

SQL

Power BI

Tableau

DOMAINS COVERED

Retail

Finance

Healthcare

Marketing

Capstone

✓ LEARNING OUTCOME

Students apply everything learned to a real, end-to-end capstone project — building a portfolio piece that demonstrates job-ready analytics skills.

Data Analytics Technology Stack

The complete toolset used across every module of the program

Programming

- Excel
- Google Sheets
- Power Query

Data Handling

- SQL
- MySQL / PostgreSQL
- Pandas

Visualization & BI

- Matplotlib / Seaborn
- Power BI
- Tableau

Reporting

- Dashboards
- Google Data Studio
- Excel Reports

Analytics & Metrics

- KPI Frameworks
- Business Metrics
- Scorecards

Domain & Capstone

- Case Studies
- Portfolio Projects
- Presentations

Hands-On Labs & Project-Based Learning

Build what you learn — every module is reinforced with hands-on work

01 Data Cleaning Lab

Clean and prepare a messy real-world dataset using Excel and SQL.

02 EDA & Visualization Project

Explore a dataset end-to-end and present findings through Matplotlib and Seaborn visuals.

03 BI Dashboard Build

Design an interactive Power BI or Tableau dashboard for a real business use case.

04 Business Reporting Lab

Build and automate a recurring business report with KPIs and an executive summary.

05 Capstone Project

Take a business problem from raw data to a final, presented analytics case study.

FINAL CAPSTONE

End-to-End Data Analytics Project

Take a business problem from raw data to a data-driven business recommendation.

Learning Outcomes

What you will be able to do by the end of the program

- ✓ Apply Excel and SQL to real business datasets
- ✓ Query and aggregate data using SQL
- ✓ Perform structured exploratory data analysis
- ✓ Create clear, effective data visualizations
- ✓ Build interactive dashboards in Power BI and Tableau
- ✓ Design and automate business reports
- ✓ Define and track KPIs and business metrics
- ✓ Apply analytics to real business domains
- ✓ Communicate insights to non-technical stakeholders
- ✓ Manage an end-to-end analytics project lifecycle
- ✓ Build a portfolio-ready capstone case study
- ✓ Prepare for data analytics job interviews

Career Opportunities

Build your career in data analytics and business intelligence

Data & Reporting

- Data Analyst
- Reporting Analyst
- Business Analyst

Business Intelligence

- BI Analyst
- BI Developer (Entry)
- Dashboard Developer

Analytics & Insights

- Analytics Consultant
- Insights Analyst
- Product Analyst

Marketing & Sales

- Marketing Analyst
- Sales Analyst
- CRM Analyst

Operations Analytics

- Operations Analyst
- Supply Chain Analyst
- Process Analyst

Advanced Careers

- Senior Data Analyst
- Analytics Manager
- Head of BI (long-term)

Career outcomes depend on individual skills, project execution, interview performance, experience level and market requirements.



WHY CHOOSE SKILL IT EDUCATION

DON'T JUST REPORT DATA. LEARN TO ACT ON IT.

Learn. Explore. Report. Decide.

SKILL IT Education builds data analysts, not just certificate holders. Through an industry-driven curriculum, hands-on labs and real BI and reporting tools, every student graduates with practical analytical skills, verified project experience and the confidence to walk into a data analytics role from day one.

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Let's turn data into decisions.

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